

AN INTERNSHIP REPORT ON

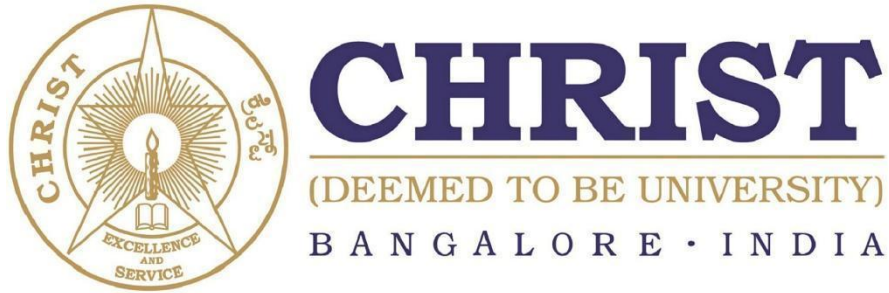
**Submitted in partial fulfillment of the requirements for the
degree of**

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
BY

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UNDER THE GUIDANCE OF
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August 2021



**Department of Computer Science and Engineering
School of Engineering and Technology
CHRIST (Deemed to be University)**

Verzeo Cyber Security Internship

by

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III Year V Sem, 2021

In partial fulfillment of the requirements of the Bachelor of Technology in Computer Science and Engineering.

Report of Internship: May 01 – June 31, 2021.
Date Submitted: 10/8/2021

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ABSTRACT

This report outlines the duties of my summer internship at Verzeo and internspool in an online medium. Verzeo is an **AI-based platform** that provides real-time insights and online training programs and internship courses for students. My internship comprised of training as well as a minor as well as a major project which was provided by internspool. This report also comprises of the activities which were performed by me in the internship period while completing my training as well as minor and major projects. The domain of my internship was Cybersecurity which is also my Honors program domain. This internship was also a team based internship where I was provided with a team in order to complete my minor as well as a major project in the guidance of my mentor who continuously guided my team during these past months and helped us in order to complete our summer internship successfully.

PREFACE

This report is prepared to fulfill the requirement of my Btech as well my honors program in cybersecurity. I completed my internship in verzeo which is an **AI-based platform** that provides real-time insights and online training programs and internship courses for students. My internship comprised of training as well as a minor as well as a major project which was provided by internspool. This report also comprises of the activities which were performed by me in the internship period while completing my training as well as minor and major projects. The domain of my internship was Cybersecurity which is also my Honors program domain. This internship was also a team based internship where I was provided with a team in order to complete my minor as well as a major project in the guidance of my mentor who continuously guided my team during these past months and helped us in order to complete our summer internship successfully. The primary focus of this report is to highlight and analyze the work done in the completion of the major and minor projects which were based on penetration testing, malware detection and analysis, password cracking and cryptography. This report highlights the tools and techniques which were used to complete the respective projects.

CHAPTER 1

1.1. Background:

Verzeo is **an AI-based platform** that provides real-time insights and online training programs and internship courses for students. Verzeo was founded in 2018. Verzeo's headquarters is located in Bangalore, Karnataka. The domain of my internship was Cybersecurity which is also my Honors program domain. This internship was also a team based internship where I was provided with a team in order to complete my minor as well as a major project in the guidance of my mentor. My minor and major projects majorly comprised of penetration testing, malware detection and analysis, password cracking and cryptography.

Penetration Testing: A penetration test, also known as a pen test, is a simulated cyber attack against our computer system to check for exploitable vulnerabilities. In the context of web application security, penetration testing is commonly used to augment a web application firewall.

Pen testing can involve the attempted breaching of any number of application systems, (e.g., application protocol interfaces (APIs), frontend/backend servers) to uncover vulnerabilities, such as unsanitized inputs that are susceptible to code injection attacks. Insights provided by the penetration test can be used to fine-tune your WAF security policies and patch detected vulnerabilities.

ii. Cryptography:

Cryptography is the study of secure communications techniques that allow only the sender and intended recipient of a message to view its contents.

1.2. Objectives:

a) To perform information gathering using tools such as Nmap, Wireshark etc.

Information Gathering: It is the act of gathering different kinds of information against the targeted victim or system. It is the first step or the beginning stage of Ethical Hacking, where the penetration testers or hackers (both black hat or white hat) performed this stage; this is a necessary and crucial step to be performed. The more the information gathered about the target, the more the probability to obtain relevant results.

b)To perform password cracking on a particular victim machine which is a part of penetration testing using tools such as crunch,hydra,john the ripper etc.

Password cracking:It is the process of using an application program to identify an unknown or forgotten password to a computer or network resource. It can also be used to help a threat actor obtain unauthorized access to resources.

c)To perform Malware analysis on a particular file given by the organization and perform static and dynamic analysis accordingly.

Malware Analysis:It is the study or process of determining the functionality, origin and potential impact of a given malware sample such as a virus, worm, trojan horse, rootkit, or backdoor.

d)To perform steganography on a given set of text which has to hidden without altering the structure of the given data.I used an open source tool called as Quickstego.

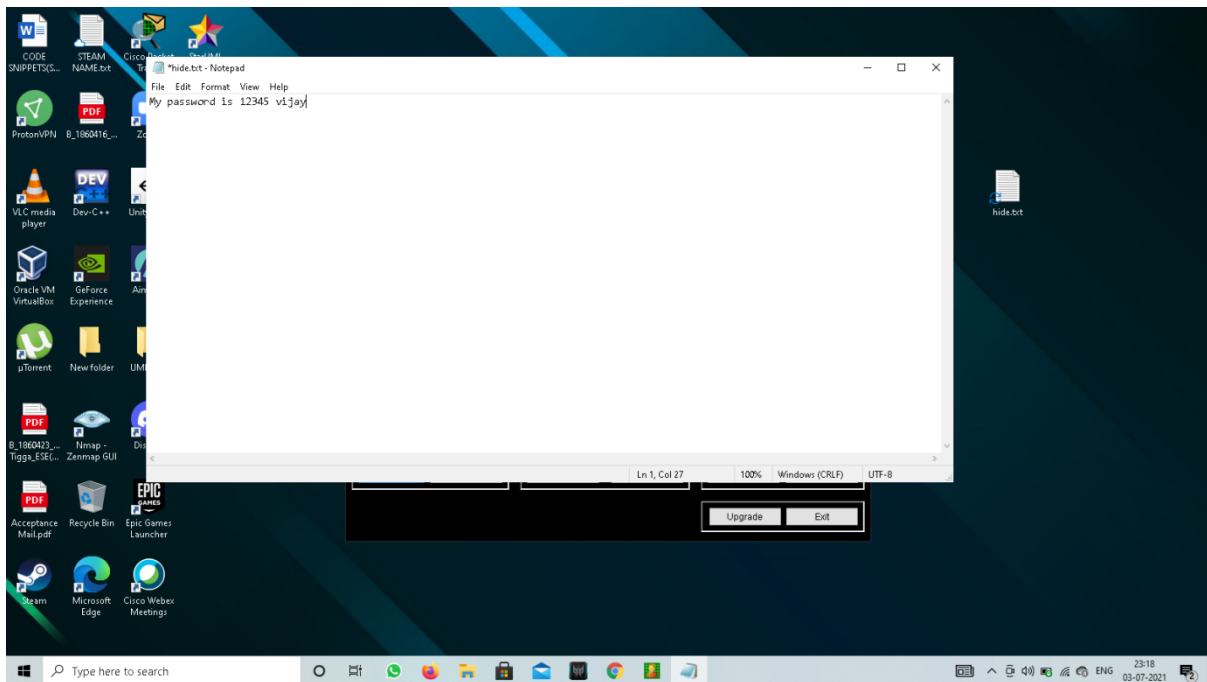
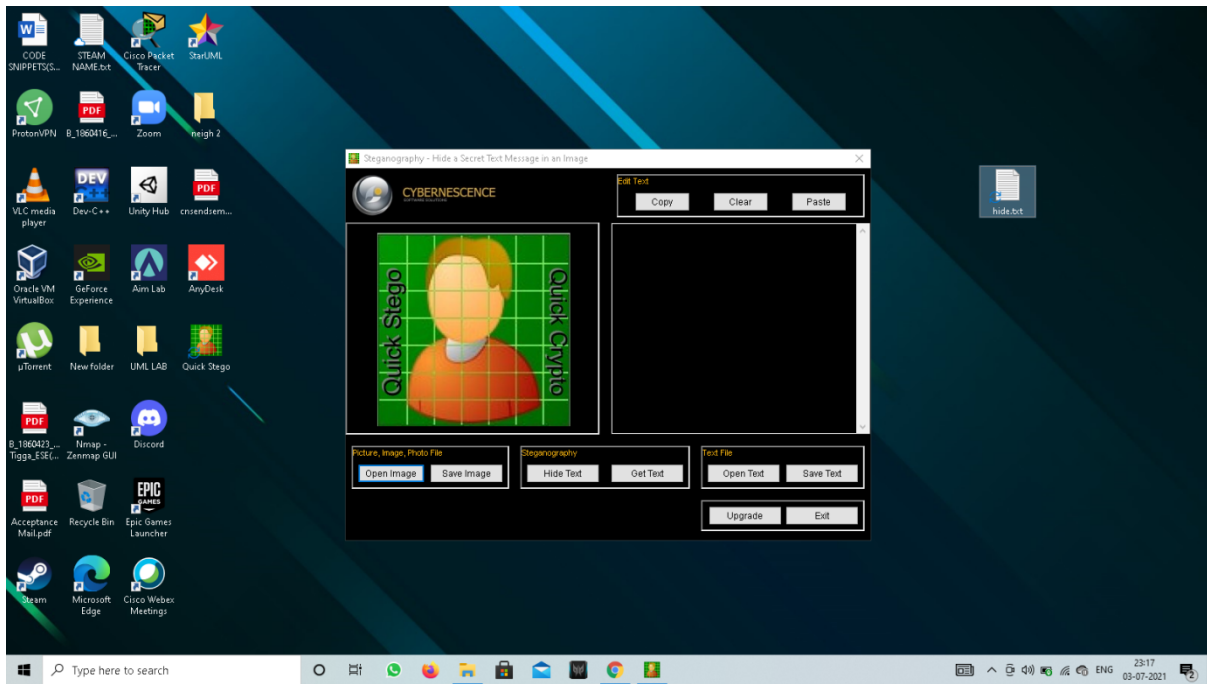
Steganography:It is the technique of hiding secret data within an ordinary, non-secret, file or message in order to avoid detection; the secret data is then extracted at its destination.

1.3.Main Contribution and steps performed in completing the minor and major project.

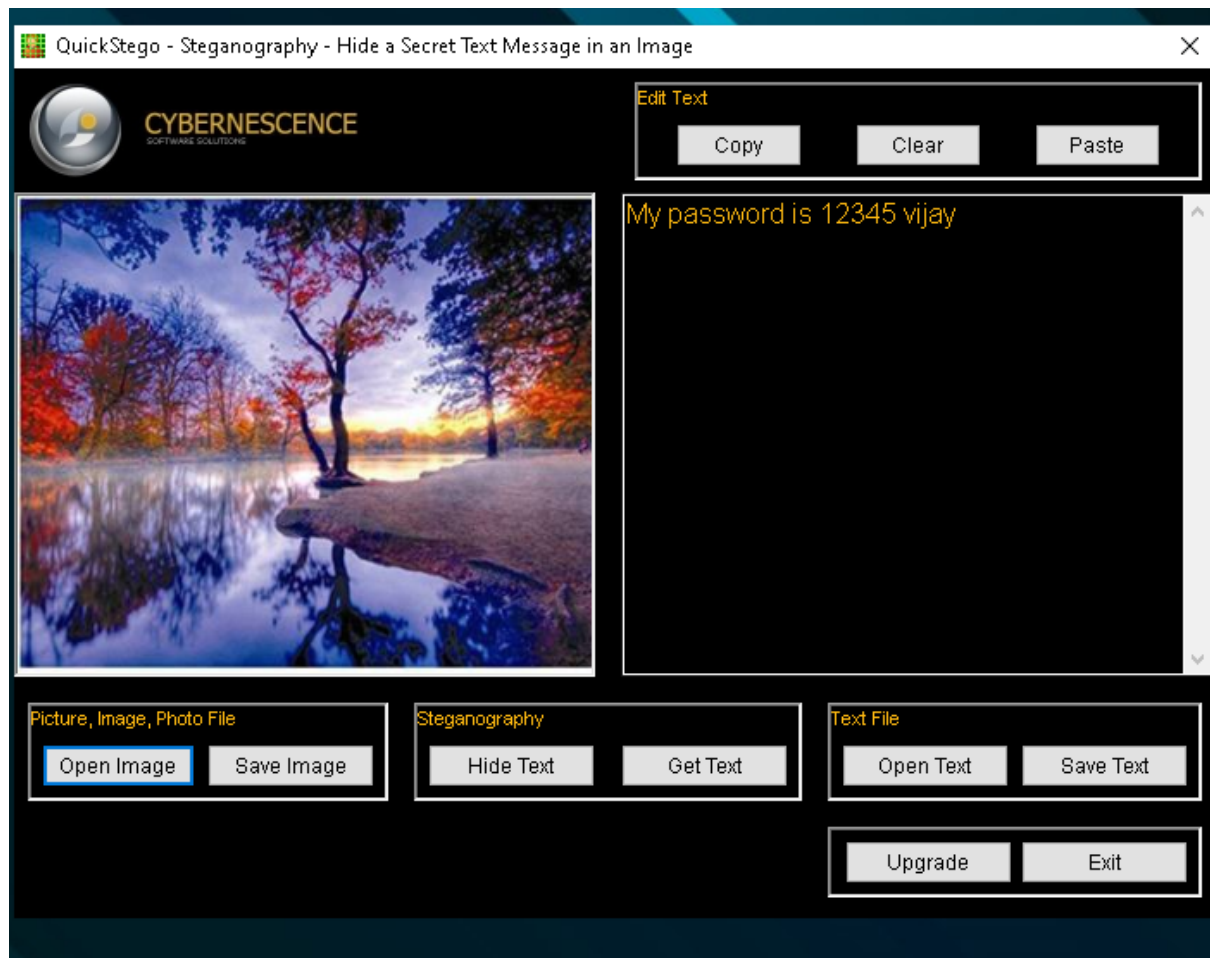
a)Minor project:My minor project was a group based project in which all of my team members including me were assigned a particular role in the project.I was given the role of performing steganography on a particular given set of text and hide it in an image for safe transmission of data.

Steps performed:

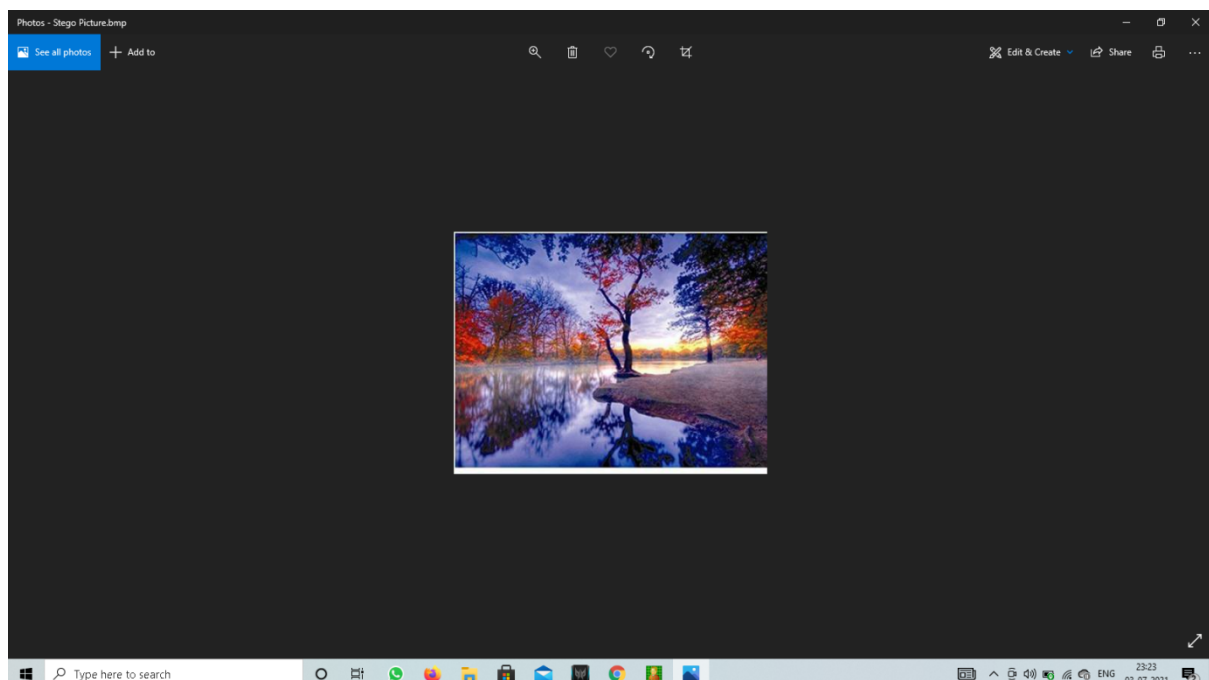
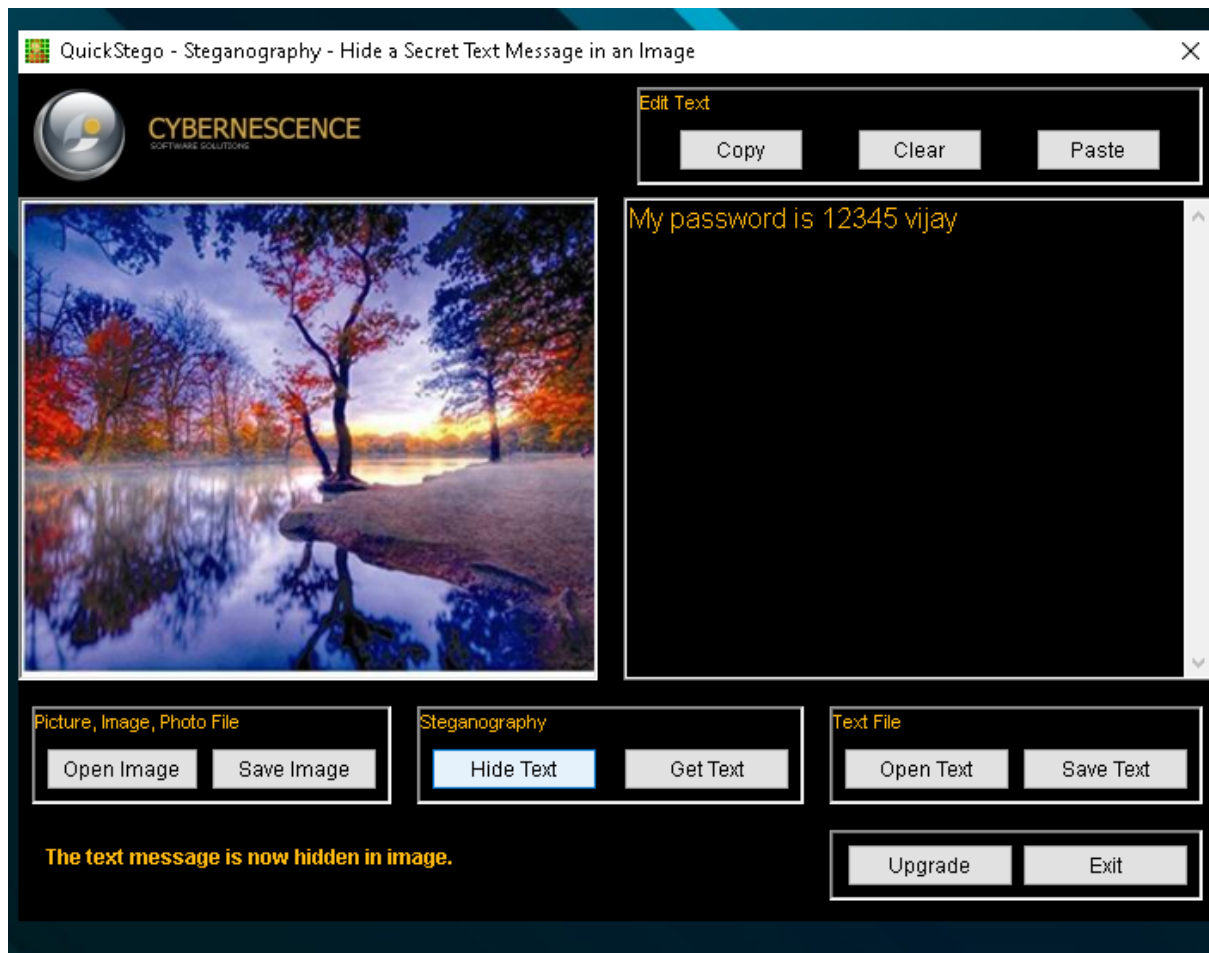
Step 1:Open quick stego tool and then create a text file containing the message to be encrypted. In my example I created a text file named hide.txt.



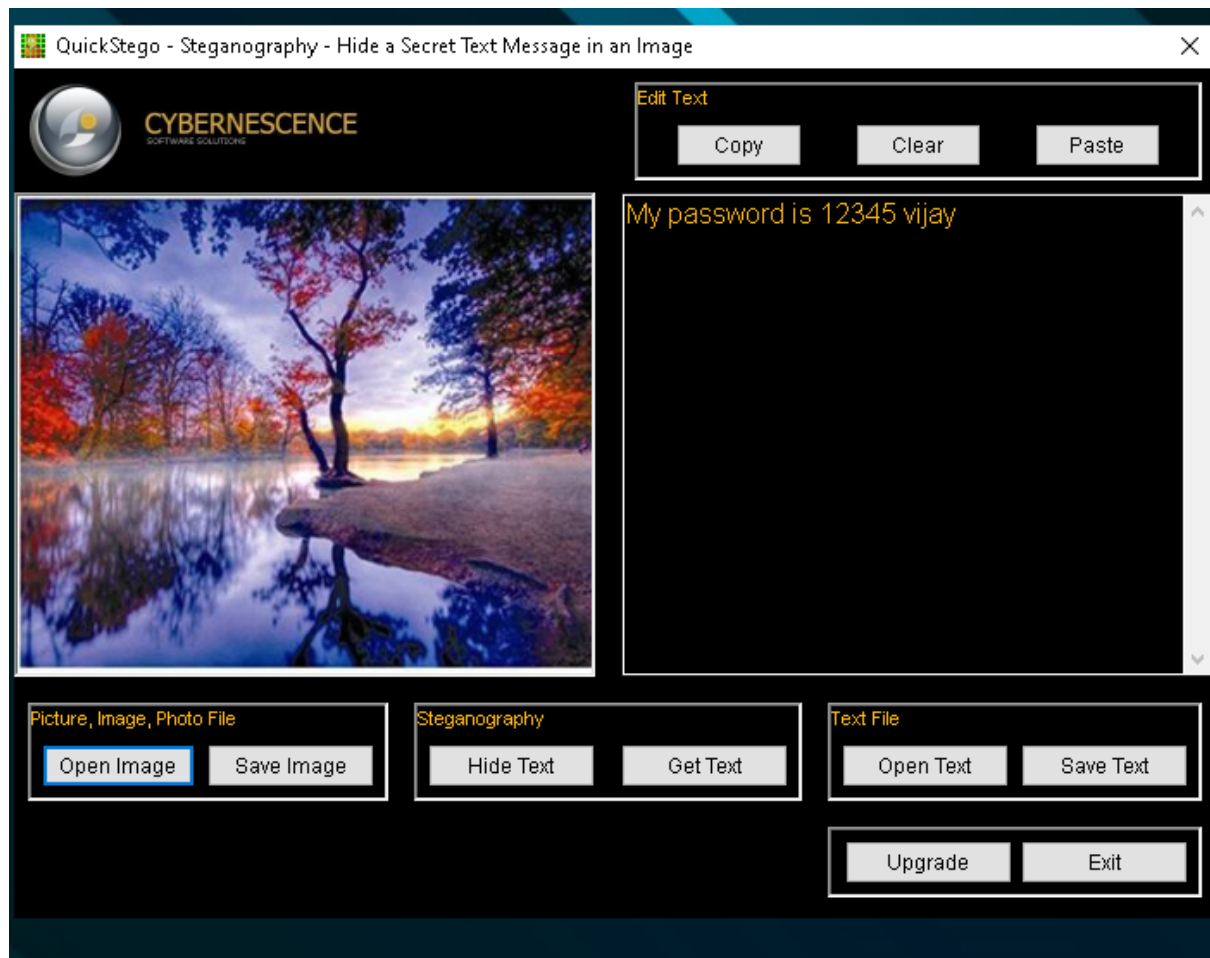
Step 2: Now open a random image in which you want to encrypt our data and open it in quick stego.



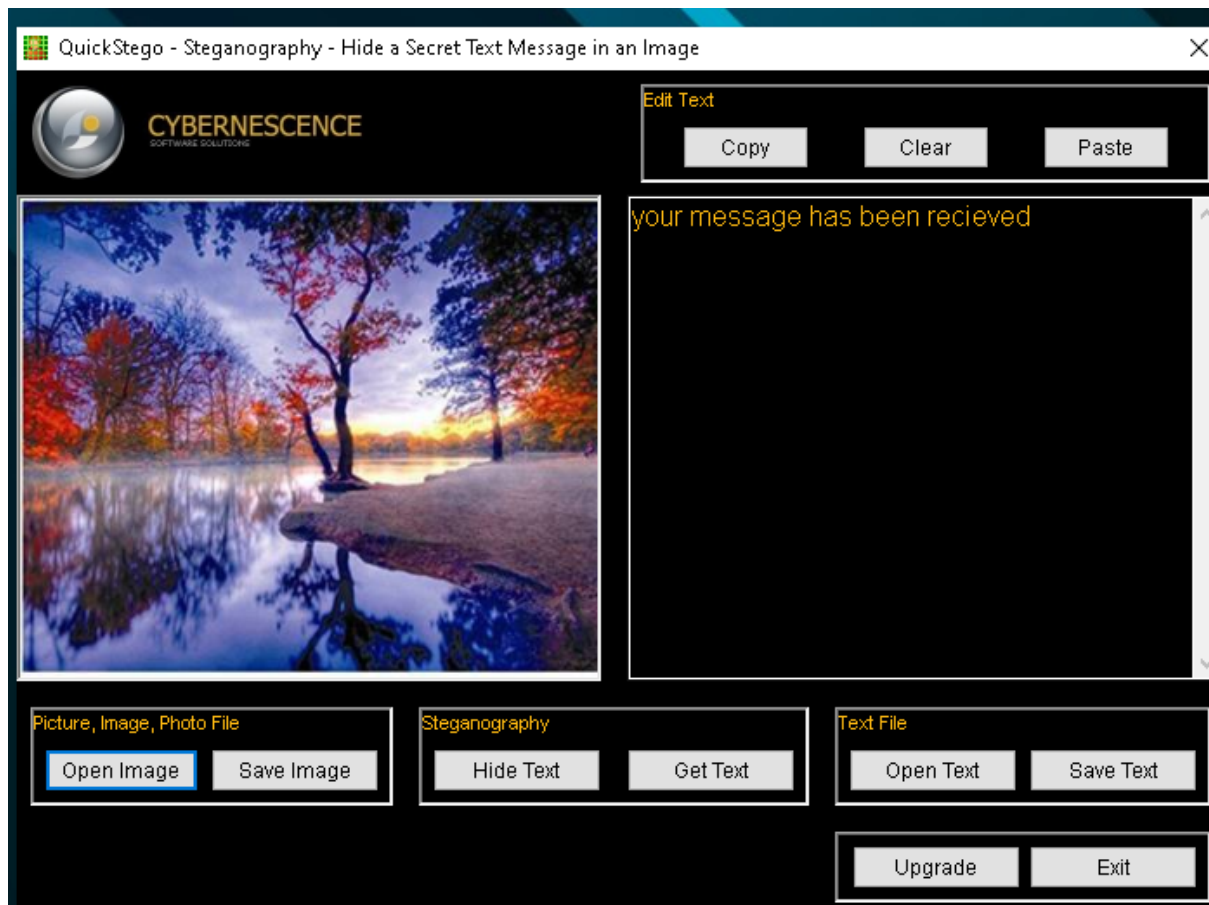
Step 3: Now open the text file we created and then press on the hide text button on quick stego tool. After that download the image which is created.



Step 4: Now to decrypt the message we need to open the image file in quick stego tool and then we would get the text which has been encrypted in the image file.



Step 5: Now we can create another text stating that the message has been received in order to notify the sender and encrypt the same into an image and send it to the sender.



b)Major project:

As Minor project,major project was also a group based project in which all of my team members including me were assigned a particular role in the project.I was given the role of performing information gathering,malware analysis,password gathering on a particular victim machine.

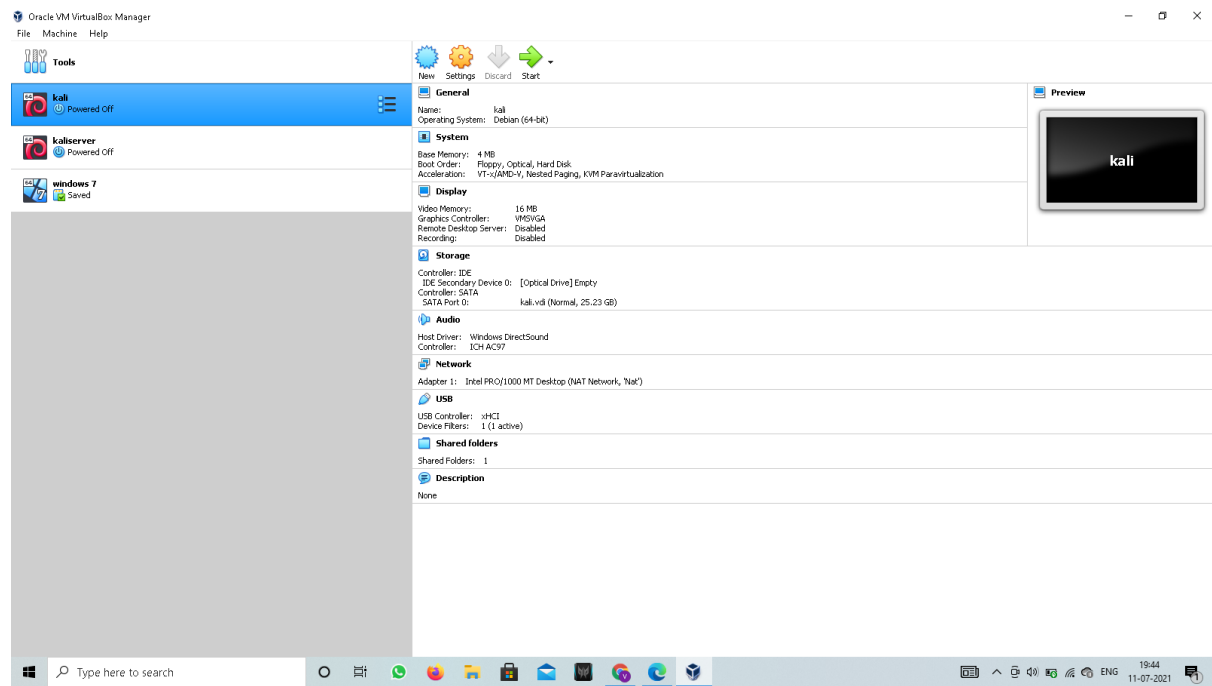
Steps performed:

1.Information gathering: Hacker Machine : Windows 10

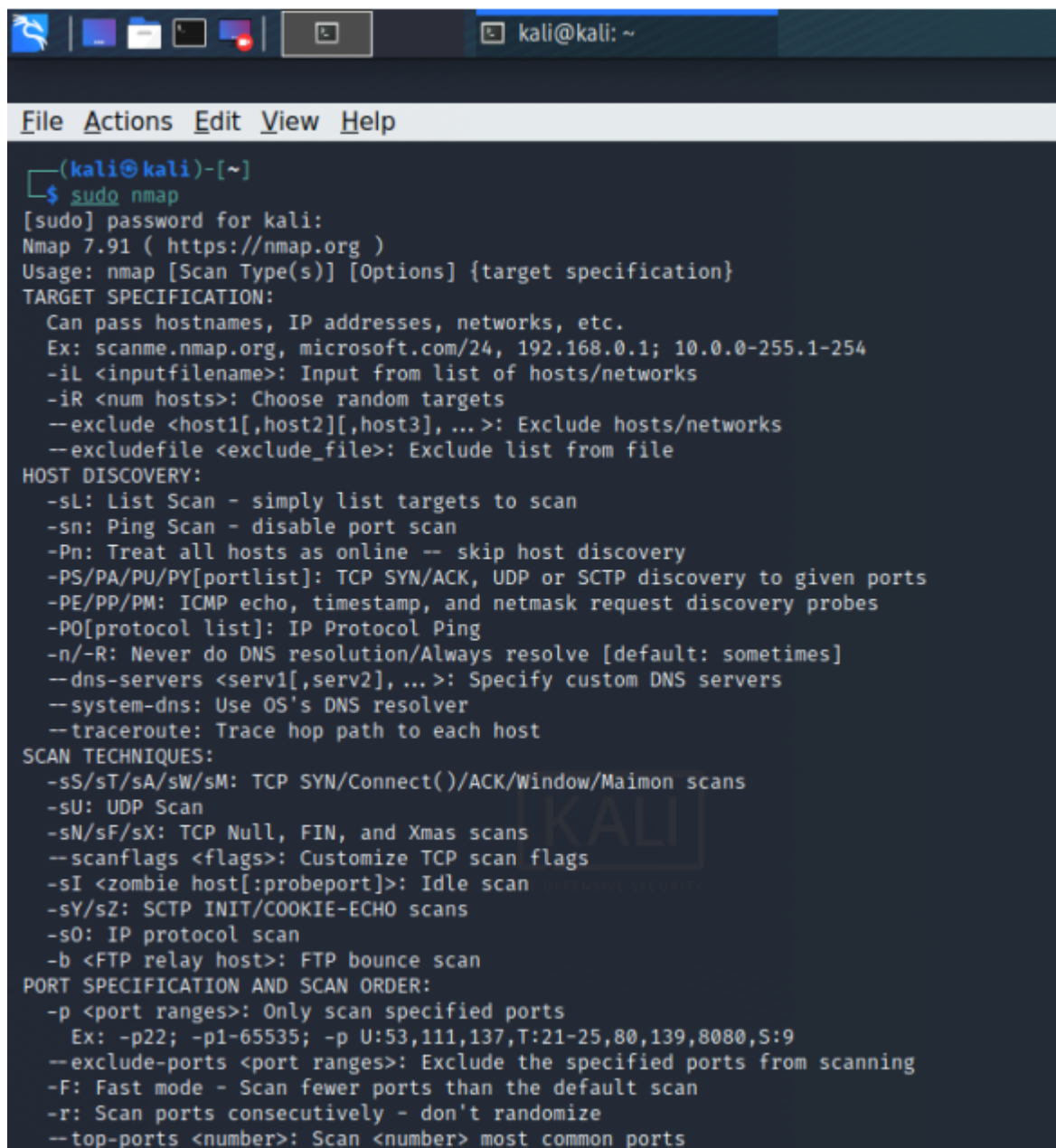
Victim machine : Kali Linux and Windows

Kali Linux::

Step 1:Open kali linux on Vm ware.



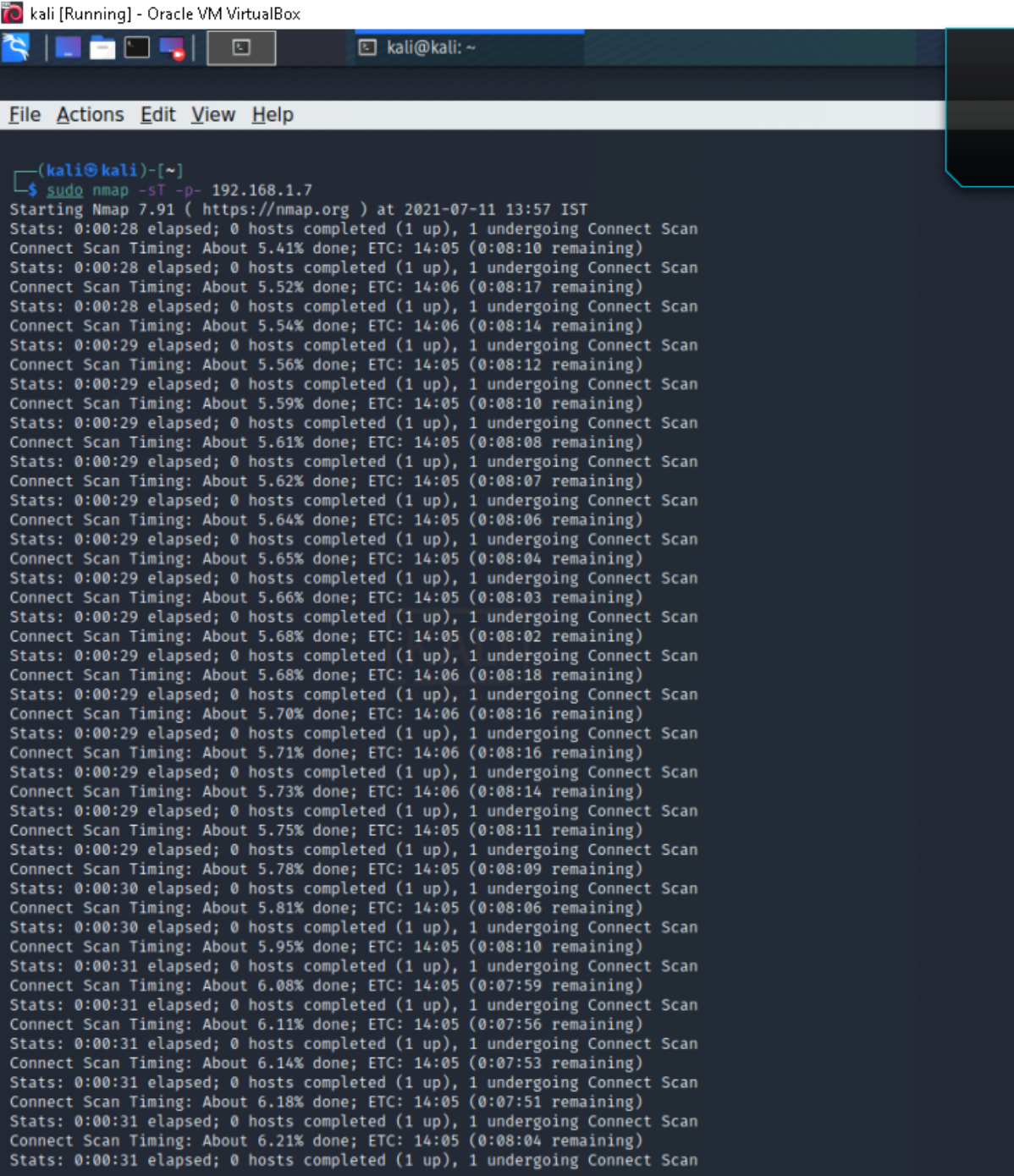
Step 2: open command terminal type the command `sudo nmap`.

A terminal window on a Kali Linux system. The window title is 'kali@kali: ~'. The terminal shows the command 'sudo nmap' being entered. The output displays the nmap 7.91 usage and various options categorized into Target Specification, Host Discovery, Scan Techniques, and Port Specification and Scan Order. A large 'KALI' watermark is visible in the background of the terminal output.

```
(kali@kali)-[~]
$ sudo nmap
[sudo] password for kali:
Nmap 7.91 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3], ...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sl: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2], ...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports consecutively - don't randomize
  --top-ports <number>: Scan <number> most common ports
```

Step3:Start the scanning process by typing the command nmap target ip.Here in this question my target ip is 192.168.1.7 and by using command sudo nmap -A 192.168.1.7 we can perform aggressive scan.Aggressive mode enables OS detection (-O), version detection (-sV), script scanning (-sC), and traceroute (--traceroute).

The sT tell nmap to scan for all Tcp open ports and -p- to scan for all 65535 ports.



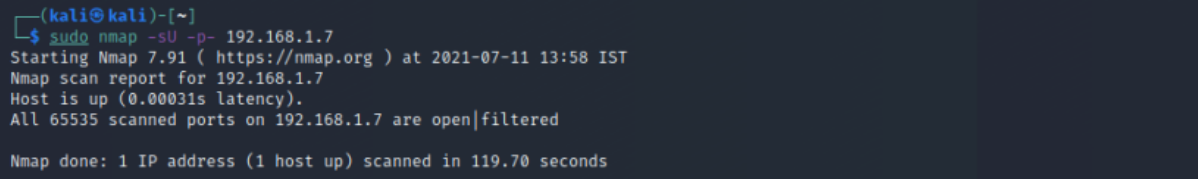
```
kali [Running] - Oracle VM VirtualBox
File Actions Edit View Help

(kali@kali)-[~]
└─$ sudo nmap -sT -p- 192.168.1.7
Starting Nmap 7.91 ( https://nmap.org ) at 2021-07-11 13:57 IST
Stats: 0:00:28 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.41% done; ETC: 14:05 (0:08:10 remaining)
Stats: 0:00:28 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.52% done; ETC: 14:06 (0:08:17 remaining)
Stats: 0:00:28 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.54% done; ETC: 14:06 (0:08:14 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.56% done; ETC: 14:05 (0:08:12 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.59% done; ETC: 14:05 (0:08:10 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.61% done; ETC: 14:05 (0:08:08 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.62% done; ETC: 14:05 (0:08:07 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.64% done; ETC: 14:05 (0:08:06 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.65% done; ETC: 14:05 (0:08:04 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.66% done; ETC: 14:05 (0:08:03 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.68% done; ETC: 14:05 (0:08:02 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.68% done; ETC: 14:06 (0:08:18 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.70% done; ETC: 14:06 (0:08:16 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.71% done; ETC: 14:06 (0:08:16 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.73% done; ETC: 14:06 (0:08:14 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.75% done; ETC: 14:05 (0:08:11 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.78% done; ETC: 14:05 (0:08:09 remaining)
Stats: 0:00:30 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.81% done; ETC: 14:05 (0:08:06 remaining)
Stats: 0:00:30 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 5.95% done; ETC: 14:05 (0:08:10 remaining)
Stats: 0:00:31 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 6.08% done; ETC: 14:05 (0:07:59 remaining)
Stats: 0:00:31 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 6.11% done; ETC: 14:05 (0:07:56 remaining)
Stats: 0:00:31 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 6.14% done; ETC: 14:05 (0:07:53 remaining)
Stats: 0:00:31 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 6.18% done; ETC: 14:05 (0:07:51 remaining)
Stats: 0:00:31 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 6.21% done; ETC: 14:05 (0:08:04 remaining)
Stats: 0:00:31 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
```

Step6: We can also perform scans for all open UDP ports.

`sudo nmap -sU -p- 192.168.1.7`

The sT tell nmap to scan for all UDp open ports and -p- to scan for all 65535 ports.



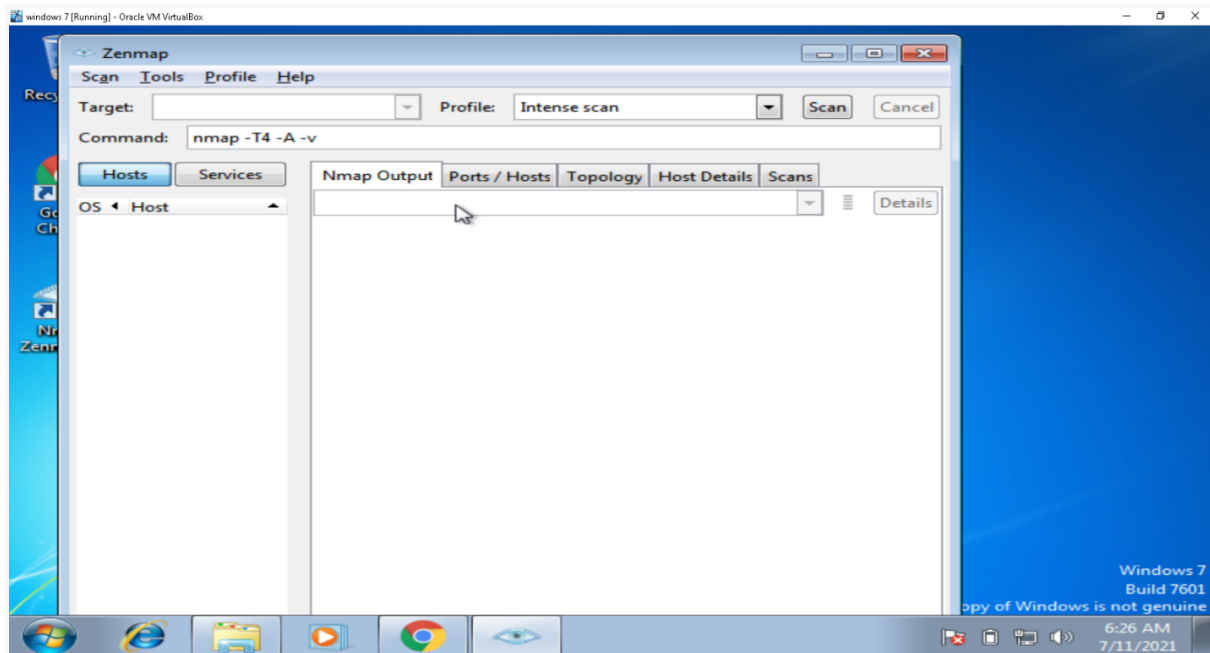
```
(kali@kali)-[~]
└─$ sudo nmap -sU -p- 192.168.1.7
Starting Nmap 7.91 ( https://nmap.org ) at 2021-07-11 13:58 IST
Nmap scan report for 192.168.1.7
Host is up (0.00031s latency).
All 65535 scanned ports on 192.168.1.7 are open|filtered

Nmap done: 1 IP address (1 host up) scanned in 119.70 seconds
```

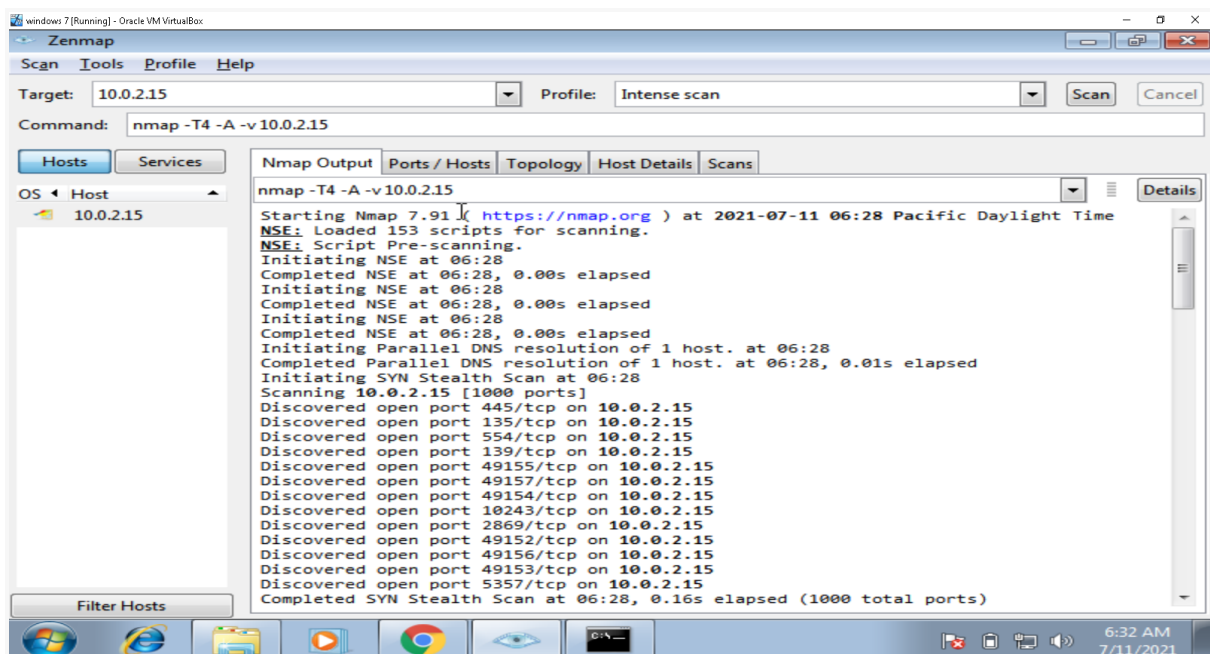
WINDOWS 7:

Target ip:10.0.2.15

Step1:Download and open nmap in windows 7 target machine.



Step2:Now just perform a normal intense scan by mentioning the target ip address.[Intense scan] command = `nmap -T4 -A -v` description = An intense, comprehensive scan. The -A option enables OS detection (-O), version detection (-sV), script scanning (-sC), and traceroute (--traceroute). Without root privileges only version detection and script scanning are run. This is considered an intrusive scan.



zenmap

Scan Tools Profile Help

Target: 10.0.2.15 Profile: Intense scan Scan Cancel

Command: nmap -T4 -A -v 10.0.2.15

Hosts Services

OS Host

10.0.2.15

Filter Hosts

Nmap Output Ports / Hosts Topology Host Details Scans

nmap -T4 -A -v 10.0.2.15

Not shown: 987 closed ports

PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds	Windows 7 Ultimate 7601 Service Pack 1 microsoft-ds (workgroup: WORKGROUP)
554/tcp	open	rtsp?	
2869/tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5357/tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
_http-server-header: Microsoft-HTTPAPI/2.0			
_http-title: Service Unavailable			
10243/tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
_http-server-header: Microsoft-HTTPAPI/2.0			
_http-title: Not Found			
49152/tcp	open	msrpc	Microsoft Windows RPC
49153/tcp	open	msrpc	Microsoft Windows RPC
49154/tcp	open	msrpc	Microsoft Windows RPC
49155/tcp	open	msrpc	Microsoft Windows RPC
49156/tcp	open	msrpc	Microsoft Windows RPC
49157/tcp	open	msrpc	Microsoft Windows RPC

Device type: general purpose

Running: Microsoft Windows 8.1|7|2008

OS_CPE: cpe:/o:microsoft:windows_8.1:r1 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_server_2008:r2

OS_details: Microsoft Windows 7 or 8.1 R1 or Server 2008 R2 SP1

Uptime guess: 0.017 days (since Sun Jul 11 06:07:13 2021)

Network Distance: 0 hops

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zenmap

Scan Tools Profile Help

Target: 10.0.2.15 Profile: Intense scan Scan Cancel

Command: nmap -T4 -A -v 10.0.2.15

Hosts Services

OS Host

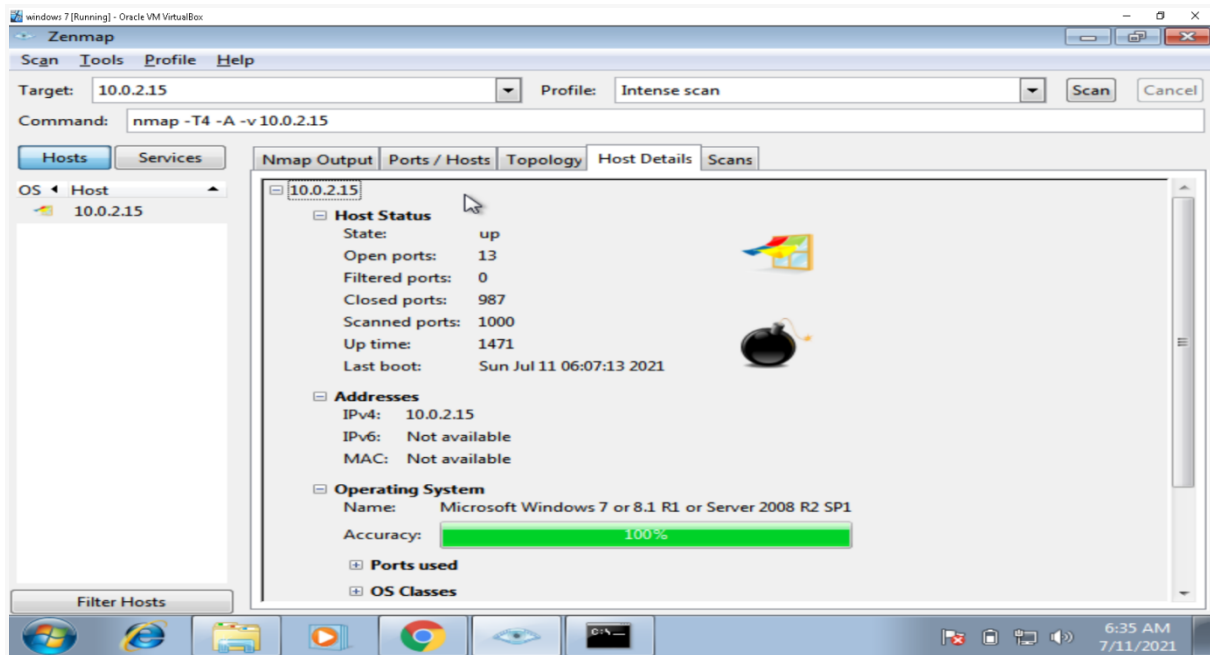
10.0.2.15

Filter Hosts

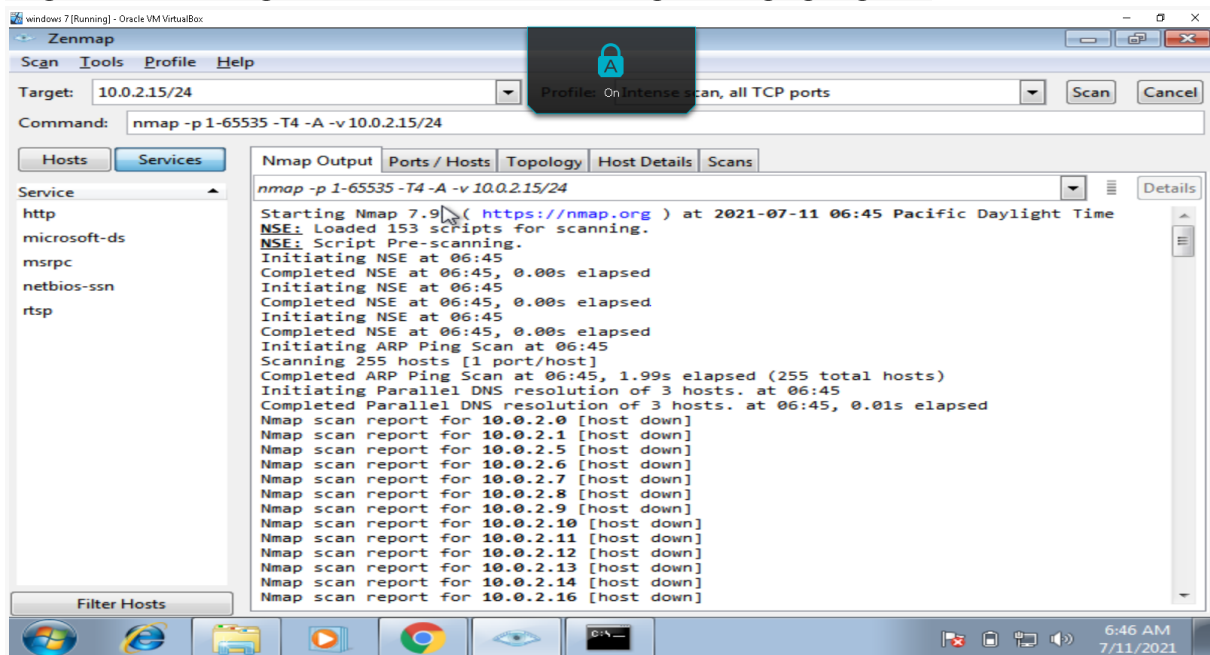
Nmap Output Ports / Hosts Topology Host Details Scans

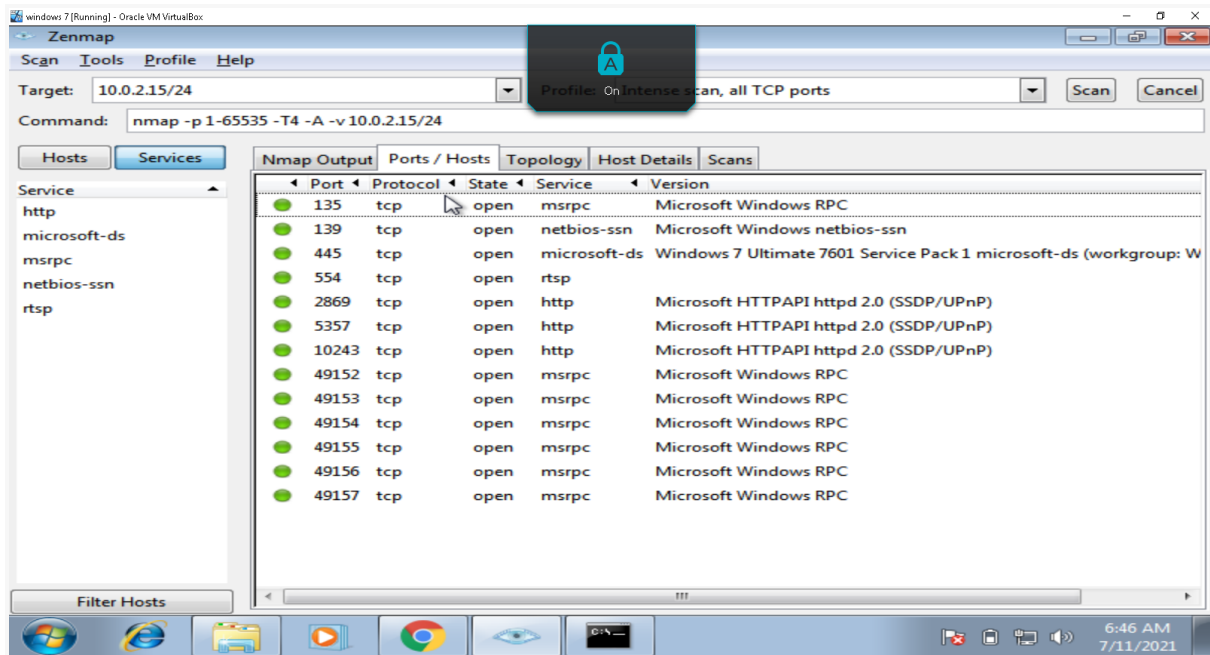
Port	Protocol	State	Service	Version
135	tcp	open	msrpc	Microsoft Windows RPC
139	tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445	tcp	open	microsoft-ds	Windows 7 Ultimate 7601 Service Pack 1 microsoft-ds (workgroup: W
554	tcp	open	rtsp	
2869	tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5357	tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
10243	tcp	open	http	Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49152	tcp	open	msrpc	Microsoft Windows RPC
49153	tcp	open	msrpc	Microsoft Windows RPC
49154	tcp	open	msrpc	Microsoft Windows RPC
49155	tcp	open	msrpc	Microsoft Windows RPC
49156	tcp	open	msrpc	Microsoft Windows RPC
49157	tcp	open	msrpc	Microsoft Windows RPC

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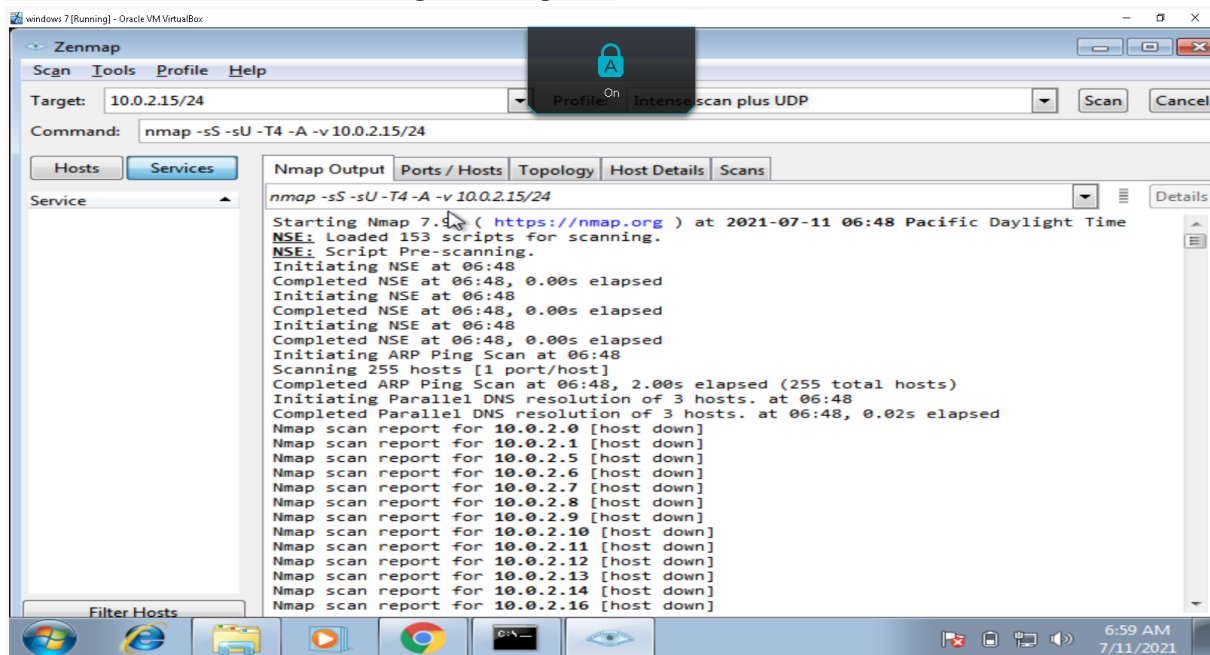


Step 3: Now we can perform intense scan for all Tcp and udp open ports:





We use -sU in order to find all open UDP ports.





We set the format as NT since Windows generally saves passwords as NTLM hashes. Wordlist contains the file which has the list of password. Passhash has the password hash which was extracted using mimikartz.

HYDRA:

First we select a target which is the ip address alternatively we can also select a list of ip addresses and also select a protocol in this case we are doing a ssh attack.

3.Malware analysis:

Steps Followed in order to complete malware analysis.

Static analysis:

Step1:First, we have to check if the malware is packed or not. For this, we will be using a trid tool, which is run using the cmd. First, we need to put the tool and malware in the same folder, and then use cmd to run the tool by going to that particular location.

Step2:To confirm if the files are packed or not, we will also be using another tool called “exeinfo PE”. Here, once we drag-drop the file into the tool, it shows all the information related to the file. At the down line, we can see that the file is not packed. Now, we will be using our next tool for static analysis.

step3:Now, we use the tool “BinText” which helps us in identifying the strings within the

malware. The strings are useful to get a rough idea about various functions the malware may be performing, and there is also a chance of getting an IP from the tool.

step 4:After this, we move to our next tool, which is “PeStudio” this tool has a lot of testing and analysing features, which can be highly useful to the tester. Here, first we go to “totalvirus” and find out the type of malware it is identified as, and also find out how many malwares are able to identify this as a malware. As we can see, most of the malwares are able to identify the malware as a trojan.

Dynamic analysis:By running the malware fille.

Next, we move to an online web application for hybrid analysis, which gives us a detailed report about the malware. It is useful to get a better idea about the malware, and the activities that the malware performs.Then we used a tool called as procmon to analyze and keep track of all the processes running in the system after running the malware file and even analyze the changes made by malware in the registry of the computer memory.

CHAPTER 2

2.1.1. Prerequisites

With the vast use of technology ,threats to system and network is increasing day by day. Cyber attackers use various techniques like malware, ransomware, phishing, and many more to compromise confidential data of large companies. Due to this, there is a tremendous demand for cyber security professionals around the world. The cyber security professional’s main job is to protect the computer and network from cyber attackers.

Some of the Prerequisites :

- A Bachelors degree in computer science engineering with minimum two to four years experience in computer network security
- Must be familiar using Kali Linux and its tools
- Must be familiar with operating systems like Unix, Linux, and Windows
- Knowledge about ISO models and cloud computing
- Up-To-Date on Relevant Technologies
- Require certification in cyber security essentials, networking and, other certified cyber courses.
- Need knowledge in Mobile forensic tools
- Should enhnace basic programming skills like java, C, C++, and PHP
- Knowledge of vulnerabilities and penetration testing

2.1.2. Responsibilities

- To monitor network and systems to detect intrusions
- Performing Cyber and Technical Threat Analysis
- Providing Timely and Relevant Security Reports
- To identify security flaws and vulnerabilities in systems and networks

- Responsible for analyzing risks and ensure the security of data
- Responsible for penetration testing
- Creating, Modifying, and Updating Intrusion Detection Systems
- To investigate cyber attacks and provide solutions to prevent them
- Record the troubleshooting

2.1.2. Challenges

With the increase of the cyber attacks, every organization needs a security analyst who makes sure that their system is secured. These security analysts face many challenges related to cyber security such as securing confidential data of government organizations, securing the private organization servers, etc.

Some challenges include :

1. Ransomware Attacks
2. Blockchain Attacks
3. IoT Threats
4. AI Expansion
5. Serverless Apps Vulnerability.

2.2. Learning Outcomes

2.2.1. Analyze and resolve security issues in networks and computer systems to secure an IT infrastructure

2.2.2. Protect data and respond to threats that occur over the Internet.

2.2.3. Design and implement risk analysis, security policies, and damage assessment

Some of the other learning outcomes include :

- Practical competence : organizational skills such as time management, dealing with systems and team work.
- Vocational competence : acquiring attitudes, behaviors, and skills related to post-college employment
- Self-awareness : self-examination
- Communication Skills : decision making, taking initiative and responsibility for one's own affairs and learning, movement from dependent to independent thinking
- Social competence : capacity for intimacy, working with others, teamwork, leadership, dealing with others, assertiveness, flexibility, public speaking, communication, patience
- Knowledge application : relating theory to practice and using skills learned in the classroom, laboratory, library, and so on in other areas of life, such as using political science theory and research methods when working in a law office
- Knowledge acquisition : academic and course-related learning, content mastery
- Academic skills : learning how to study, to write, to conduct independent research

CHAPTER 3

3.1)Conclusion:

This internship has been an excellent and rewarding experience. I can conclude that there have been a lot I've learnt from my project work at Verzeo. Needless to say, the technical aspects of the work I've done are not flawless and could be improved provided enough time. As someone with no prior experience with any project and industrial work experience whatsoever I believe my time spent in it was well worth project work and contributed to finding an acceptable solution to build a fully functional web service. Two main things that I've learned the importance of are time-management skills and self-motivation.

Finally, my time at Verzeo helped me to develop a basic understanding on the various Cyber Security concepts. Overall, this was a very good learning experience.

3.2)Future Scope:

This internship at Verzeo helped me to gain first-hand exposure of working in the real world. It also allowed me to harness the skill, knowledge, and theoretical practice they learned in university. This Internship provided a nice learning curve for me with little experience of the professional world. This experience will help me to gain a job in the cyber security domain in the future and even help me to build future career possibilities in the field of cyber security.

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